

hw2

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1 Biomedical Data Science - Assignment 2

Shane Ferrell

1.1 1) Sorting Hat Program

```
[1]: def sorting_hat():  
  
    gryffindor_score = 0  
    slytherine_score = 0  
  
    answer = input("Is a hot dog a sandwich? (Y/N): ")  
    if answer == "Y":  
        gryffindor_score = gryffindor_score + 1  
    elif answer == "N":  
        slytherine_score = slytherine_score + 1  
  
    answer = input("Does pineapple go on pizza? (Y/N): ")  
    if answer == "Y":  
        gryffindor_score = gryffindor_score + 1  
    elif answer == "N":  
        slytherine_score = slytherine_score + 1  
  
    answer = input("Would you choose to fight one horse-sized duck over 100_  
↳duck-sized horses? (Y/N): ")  
    if answer == "Y":  
        gryffindor_score = gryffindor_score + 1  
    elif answer == "N":  
        slytherine_score = slytherine_score + 1  
  
    return "Gryffindor" if gryffindor_score >= slytherine_score else_  
↳"Slytherine"
```

1.2 2) mydata Class

```
[2]: class mydata:

    def __init__(self, numbers):
        self.numbers = numbers

    def mean(self):
        return sum(self.numbers) / len(self.numbers)

    def std(self):
        return (sum((x - self.mean()) ** 2 for x in self.numbers) / (len(self.
↪numbers))) ** 0.5
```

```
[3]: x = mydata([1, 2, 7, 5])
print(x.mean())
print(x.std())
```

```
3.75
2.384848003542364
```

1.3 3) lag Function

```
[4]: import numpy as np
```

```
[5]: def lag(a, n):
    return [np.nan] * n + a[0:-n]
```

```
[6]: a = [1, 6, 8, 4]
print(lag(a, 1))
print(lag(a, 2))
```

```
[nan, 1, 6, 8]
[nan, nan, 1, 6]
```

1.4 4) Lag Generator

```
[7]: def lagGenerator(n):
    return lambda a: lag(a, n)
```

```
[8]: lag_2 = lagGenerator(2)
lag_2(a)
```

```
[8]: [nan, nan, 1, 6]
```